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BlockDrugChain: A blockchain and ML-driven framework for scalable drug discovery management

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ABSTRACT

The discovery of new drugs is a collaborative effort that involves researchers, pharmaceutical companies, and regulators. Cyberattacks, data tampering, and a lack of transparency are all common problems in today's centralised systems. This paper suggests using Hyperledger Fabric-based blockchain technology to increase the efficiency, transparency, and safety of drug discovery. The straightforward user interface makes it easy to upload, update, and validate data. Each contributor is granted a unique ID using SHA-256, which protects the data. With the use of machine learning, data is improved and people are empowered to make better decisions. To expedite verification, critical metadata is stored on-chain, while large files are stored off-chain. Custom smart contracts safeguard your data and transfer ownership to you. According to performance studies, it can process up to 365 transactions per second, making it ideal for large-scale collaboration. An alternative to traditional drug discovery approaches that is both modern and decentralised is offered by this idea.

1. Introduction

Novel drug development involves several stages, including preclinical research, target identification, chemical synthesis, analysis of genetic data, and stratified human clinical trials [1]. No amount of scientific brilliance will be enough to solve the problems of trust, openness, data security, and cooperation. Concerns regarding operational issues, unauthorised data modifications and breaches have been voiced by Bahri et al. [2] regarding centralised systems inside research institutes, pharmaceutical enterprises, and regulatory organisations. Data breaches and ransomware are becoming more commonplace for pharmaceutical and healthcare companies. Data integrity, access control, and traceability are essential components of any decentralised system [3]. Blockchain technology is poised to transform critical industries including healthcare and pharmaceutical research. Chintala [4] argues that it has the ability to increase trust, transparency, and permanence. With blockchain technology, there is no longer any room for error because every transaction is recorded in a decentralised ledger that can be easily checked. Confidential information can be sent securely by reliable parties. Consider Hyperledger Fabric and similar permissioned blockchains if your company places a premium on privacy, security, and control over user access. Debe et al. [5] note that in order to accomplish scalability and regulatory compliance, they employ certificate-based identity management, encrypted data transmission, and consensus validation. These days, pharmaceutical R&D would be incomplete without machine learning. According to Hunge et al. [8], it facilitates the process of discovering targets, screening chemicals, evaluating toxicity, and optimising compounds. Traditional wisdom and molecular patterns in biological data can both be enriched by machine learning methods. When combined, blockchain technology, artificial intelligence, data provenance, and machine learning generate trustworthy and safe insights. The use of AI has transformed healthcare diagnostics, personalised treatment plans, and illness prognosis. "Artificial Intelligence in Drug Discovery and Development Against Antimicrobial Resistance" and "Haematology and Hematopathology Insights Powered by Machine Learning: A Narrative Review on Shaping the Future of Blood Disorder Management" reveal how AI has the potential to improve biological operations. "Effective Text Classification Using BERT, MTM, LSTM, and DT" demonstrates how AI can adapt to medical data that is not well-formatted. This facilitates therapy and research. Based on these results, it seems that AI and blockchain can be integrated to build drug development ecosystems that are smart, secure, and open.

A drug discovery strategy that combines blockchain technology with machine intelligence is introduced in this paper as BlockDrugChain.

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Drug research data may be securely uploaded, edited, and validated using Hyperledger Fabric and an intuitive DApp. Data in Hyperledger Fabric is protected by SHA-256 hashing and X.509 certificate-based identity management. The requirement that access and authority be allowed exclusively to permitted users is imposed by G. K. Jayaraman et al. [6]. To reduce blockchain costs, huge datasets are stored off-chain, while only metadata and InChIKeys are kept on-chain [7]. Hunge et al. [8] states that early drug development decision-making is improved by using machine learning for data preparation and target quality assessments.

1.1. Motivation

Conventional drug discovery techniques are prone to mistakes, delays, and data breaches since they depend on centralised and heterogeneous data systems. There is a pressing need for a platform that guarantees:

√ Trust-based collaboration.
√ Unbreakable pharmaceutical lifecycle contribution monitoring.
√ Protecting data flow, auditability, and privacy.
√ Faster data ownership and source verification.

BlockDrugChain aims to solve these issues by building a safe, open, and scalable decentralised drug discovery ecosystem [9]. Blockchain and machine learning provide analytical and predictive characteristics that improve early-stage drug candidate selection.

1.2. Research gaps

Few studies have examined blockchain's overall function throughout the drug development process, despite the fact that several have examined its potential uses in supply chains and healthcare. The following major issues have been uncovered by this analysis:

√ Machine learning modules that support data analytics and visualisation are lacking in the field of blockchain-based pharmaceutical development.
√ From managing clinical trials to integrating genetic data, problems arise from the absence of a complete, scalable decentralised application.
√ lack of infrastructure for solutions for off-chain storage that improve scalability and make data tracing easier.
√ Insufficient identity management and certification system installation to guarantee accountability and verify contributions.
√ Inadequate performance metrics to assess scalability in relation to various transactional and organisational needs.

BlockDrugChain's multi-tiered, permissioned architecture aims to directly address these issues.

1.3. Novelty of the approach

BlockDrugChain is making a novel platform for drug discovery management that is flexible and scalable while keeping users' privacy. It does this by combining blockchain technology with machine intelligence.

√ The execute-order-validate architecture of Hyperledger Fabric makes it much more popular because it allows for high throughput and low latency performance.
√ Using SHA-256 and X.509 certificates makes sure that transactions are safe and that people are who they say they are.
√ Complex chaincode modules were made to handle uploading, modifying, validating, and issuing certificates in real time.
√ Machine learning was used to preprocess and visualise the target data from the ChEMBL database in order to make early candidate screening better.
√ Our tiered system architecture combines people, applications, interfaces, analytics, smart contracts, and blockchain to make sure that everything works as well as possible and can be changed as needed.
√ The front-end DApp interface of our whole system makes it easier for regulators, contributors, and reviewers to talk to each other.
√ The Calliper benchmarking tool showed that changing endorsers and block size made the network work better.

1.4. Key contributions

This work contributes significantly to the field of safe and scalable drug development.

√ Developed and implemented a decentralised blockchain (BlockDrugChain) to monitor drug discovery contributions throughout the lifecycle.
√ Creating ML modules using ChEMBL genomic data to assess target quality.
√ Using Hyperledger Fabric with SHA-256 unique IDs and certificate-based access control ensures secure asset management.
√ Use smart contracts (chaincode) to enrol people, handle contributions, and issue certificates.
√ Off-chain storage of contribution data and on-chain metadata allows for greater scalability.
√ End-to-end DApp with user interface and API endpoints for secure and interactive blockchain access.
√ Performance testing demonstrates improved throughput (up to 365.2 TPS) and scalability across several companies.
√ A comparison of how the new blockchain model performs better for pharmaceutical use cases than older ones.

2. Literature survey

Several data-intensive steps comprise the lengthy and intricate drug discovery pipeline, such as genomic analysis, target identification, medication screening, and clinical trials Mahameed, I., et al. [11]. During these stages, there must be secure collaboration among research institutions, pharmaceutical businesses, biotech firms, and regulatory bodies. The use of centralised data storage and management systems has its limitations, such as limited interoperability, the possibility of data tampering, and the introduction of single points of failure Mallavarapu, J. R., et al. [12]. Because modern pharmacological research is ever more complicated, we need to use digital technologies that are safer, more open, and easier to scale. New developments in blockchain and machine learning (ML) provide exciting chances to change the way drug research is done.

2.1. The impact of blockchain on healthcare and pharmaceutical research

Blockchain, or distributed ledger technology, guarantees data accuracy, immutability, and traceability. Blockchain technology has shown promise in three areas of healthcare management: electronic health records, medical supply networks, and methods for getting patient consent. Mohammed Talib & Associates [13]. Extensive research has been conducted on the potential applications of blockchain technology inside the pharmaceutical supply chain. Nageshwaran, V [14] illustrated the application of blockchain technology to monitor pharmaceuticals along the supply chain, hence mitigating occurrences of medicine counterfeiting. Palli, P., and S. Mishra [15] established a medical supply chain utilising blockchain technology to ensure product legality and traceability. Rather than rethinking the entire process of pharmaceutical production, most of these efforts focused on improving logistics and the supply chain. According to Prabhu, M. et al. [16], by permanently

preserving techniques and outcomes, blockchain can improve clinical study repeatability and transparency. Despite these developments, there are still a few comprehensive blockchain systems that can perform basic operations like drug candidate screening and genetic data management.

2.2. The current status of drug discovery systems and their limitations

Traditional drug development approaches rely heavily on centralised data storage.

- Data breaches and privacy violations occur, especially in corporate collaborations. G. Yueli et al. [25].
- There are insufficient resources to authenticate data and ensure conformity to regulations. In 2024, Rao et al.
- In the absence of specific control of contributions, it is difficult to determine intellectual property ownership.
- Scalability concerns develop as a result of restrictions in centralised storage and computing.

The absence of automated access control measures and tamper-proof audit trails precludes present systems from fully supporting multi-party drug discovery operations.

2.3. Hyperledger fabric for consortium use

Hyperledger Fabric is a permissioned blockchain platform intended specifically for enterprise use. Collaborative applications, such as the discovery of novel medications, are particularly suited to this. Ethereum and Bitcoin are examples of public blockchains, but Hyperledger Fabric enables private transactions, provides adjustable consensus algorithms, and imposes strict access controls. This is the best solution for high-volume transaction circumstances, such as pharmaceutical R&D, because of its unique execute-order-validate methodology, which enhances throughput while lowering latency. Saha, R., et al. [18] created Hyperledger Fabric to help institutions share medical records. MedRec, a blockchain-based medical record system established in 2016, has not addressed the issue of medication discovery and does not allow for the integration of machine learning. Sayal A, et al. [19]. These scenarios do not address the collaboration to develop new drugs, but they do highlight how Fabric can safely manage healthcare data.

2.4. The function of machine learning in drug discovery

Machine learning has emerged as an essential element of modern drug discovery by automating and enhancing many tasks [10].

√ Employing bioactivity databases like ChEMBL to predict molecular activity.
√ Employ both supervised and unsupervised learning to discover target proteins.
√ Extensive biological data feature extraction and dimensionality reduction.
√ Utilisation of deep learning models for medication repurposing and candidate evaluation.

DeepChem, ChemBERTa, and Mol2Vec exemplify machine learning technologies that have accelerated compound screening and prediction processes. Notwithstanding its use, machine learning systems frequently lack a secure and verifiable architecture to safeguard training data and outputs, rendering them susceptible to manipulation or unauthorised access in multi-party environments Vempaty, et al. [23].

2.5. Integration of blockchain and machine learning

Researchers are looking into how to combine blockchain with machine learning to get reliable AI-driven insights. Siradanai, et al. [20] came up with a blockchain-based federated learning architecture to move medical data. Unfortunately, the model could only handle EHRs. Suri, A., et al. [21] looked into the possibility of using blockchain technology to store hashes of machine learning models and make sure that these models are safe. However, they did not specifically look at how this may be used in the pharmaceutical industry. This opens up a chance for future research: developing hybrid frameworks that use machine learning to make the system better at analysing data while keeping the data and models safe. To get over these problems, this work introduces BlockDrugChain, a decentralised program for managing drug discovery in a safe and scalable way V, S., R, et al. [22]. It uses Hyperledger Fabric and has machine learning built in The proposed system combines cryptographic assurance, fine-grained access control, smart contracts, off-chain storage, and machine learning-driven analytics to create a complete solution for modern pharmaceutical research and development. It ensures privacy, scalability, traceability, and compliance with the law, and it lets various stakeholders work together on data in real time.

2.6. Research gaps

Academics are starting to pay more attention to blockchain-enabled healthcare systems, but current research shows that these platforms aren't up to the task of monitoring the whole drug discovery process due to technological flaws. Machine learning and blockchain are not integrated into most systems' designs. Data tracking and analysis becomes more complicated as a result. Storing and expanding huge molecular or clinical datasets is made more difficult by the fact that current models do not provide off-chain data management capabilities. Also, different companies have not set up identity and verification processes to check if contributions are real and responsible. Research on scalability and performance benchmarks within the context of real organisational demands is limited. This suggests that these systems' resilience has not been assessed. Validation tools that enable stakeholders to collaborate are often absent from existing implementations, and interactive user interfaces are also lacking. To solve these problems, BlockDrugChain designed a modular architecture that incorporates machine learning analytics with blockchain technology, secure off-chain storage, comprehensive benchmarking, a decentralised user interface, and certificate-based identity management. This makes drug discovery operations scalable, transparent, and secure.

3. Proposed approach

BlockDrugChain uses Hyperledger Fabric to clean and analyse data using blockchain technology and machine learning. This allows for the decentralised, secure, and scalable control of medication development. Data integrity breaches, unauthorised access, interrupted collaboration, and inadequate verifiable traceability are some of the problems that could arise from using antiquated approaches in drug discovery. That is the goal of this approach.

3.1. A safe method for discovering new pharmaceuticals using blockchain technology

To find new medicines, a long process of difficult tasks must be completed. Genomic data sequencing, protein target identification, chemical synthesis, animal testing, and multi-phase clinical trial design are all part of this process. Data created during these times is often scattered, vulnerable to modification, or compromised during breakdowns in outdated systems. Our method turns these operations into contributions that cannot be changed and are instead recorded on the blockchain in an immutable form. For unique identifiers, SHA-256 encrypts every data segment. Finding it and showing that it hasn't changed is now possible with this. When it comes to business applications, permissioned blockchain technology Hyperledger Fabric shines. To handle

these payments, we built a decentralised application (DApp) using it. Fig. 1 shows the Secure Drug Discovery Chain Process Backed by Blockchain Technology. Ensuring that only verified and approved individuals may see, post, edit, or authorise contributors is its primary goal. This is done by a Certificate Authority (CA) digital certificate that is X.509 compliant. A contributor-initiated transaction must always follow the strict protocol of "execute, order, and validate." To ensure the logic is correct before adding it to the ledger, peers use chaincode, which are smart contracts. After their names have been confirmed, this takes place. The legitimacy of recorded gifts and the legal responsibility of the donor can be confirmed by any member of the network.

3.2. Role-Based access and contribution validation

Hospitals, research labs, biotech firms, and government organisations can collaborate independently on the Hyperledger consortium N

Nnetwork. After registering, users can choose to receive an online orientation. Once the administrator approves, each user is given a digital certificate and rights according to their job. The duties that follow are as follows:

√ For users (researchers, analysts), posting and keeping contributions to drug development is crucial.
√ Administrators (Regulatory Authorities): Manage who can authorise donations, grant privileges, and access what.
√ Coworkers: Oversee the database and its modifications.
√ Endorsers: Before blocks are created, confirm that smart contracts are being executed.

We authenticate all transactions for user credentials and permissions before to accepting them in order to prevent unauthorised access and confirm the legitimacy of donations. By keeping data private and verifiable, this approach encourages accountability and transparency among participants.

3.3. A layered architecture for adaptable and modular design

Different functions are handled by the highly recommended six-tiered system. When combined, these layers enhance the performance, security, and scalability of real-time systems. Fig. 2 illustrates how a complex Layered Architecture makes the proposed approach adaptable and expandable.

Those who work in the fields of genetics, target analysis, animal testing, and clinical trials are included in the first level, known as the

1. User Layer. Every user has a distinct digital identity that they use to access the system. The DApp's online interface is made to be easy to use.
2. Application Layer: The Application Layer offers several features. It facilitates the submission, editing, verification, and issuance of contribution certificates.
3. Interface Layer: Communication between the application and the blockchain network is made possible via the Interface Layer. This includes the backend logic and RESTful APIs. They provide both blockchain-based transactions and off-chain data enquiries.
4. Analytics Layer: Uses machine learning algorithms to preprocess and show data related to medications. Researchers use ChEMBL bioactivity data to determine possible protein targets, for example. Using ML algorithms, the system calculates quality scores for each protein and shows them to researchers. This layer is helpful for making choices about chemicals early on.
4. Smart Contracts Layer: This layer uses chaincode to wrap up all of the business functions, such as verifying data, uploading it, making certificates, controlling access, and so on. It makes sure that all DApp actions have a safe and verifiable logic flow.
6. Blockchain Layer: The blockchain layer is in charge of keeping track of all transactions and their information. The genesis block holds the initial configuration data, and new contributions are added to the chain as blocks.

3.4. Hybrid on-chain and off-chain storage mechanism

Due to the large size and complexity of biomedical datasets, including molecular structures, preclinical test reports, and clinical trial records, the direct storage of comprehensive data objects on the blockchain would result in performance and scalability issues. The BlockDrugChain framework implements a hybrid storage strategy that efficiently balances security, traceability, and system efficiency in order to resolve this issue. Only the most critical metadata, including the SHA-256 hash of the contribution, the timestamp, the InChIKey (chemical structure identifier), and the contributor identity, are stored on-chain in this model. This guarantees the integrity and immutability of critical identifiers, while simultaneously facilitating rapid verification. In the interim, the blockchain's storage and computational burden are mitigated by the secure preservation of the datasets in off-chain storage. As more organisations and contributions join, this architecture makes sure

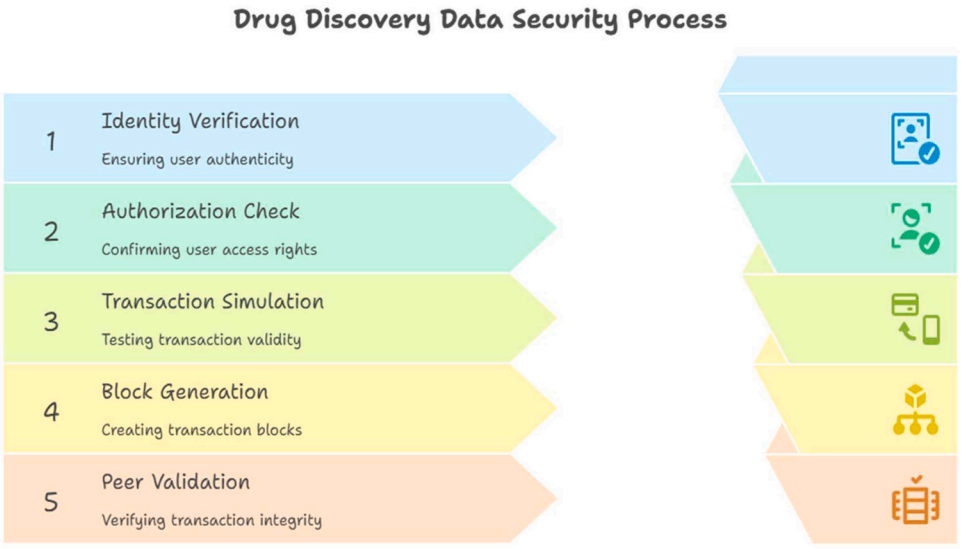


Fig. 1. Blockchain-backed secure drug discovery chain process.

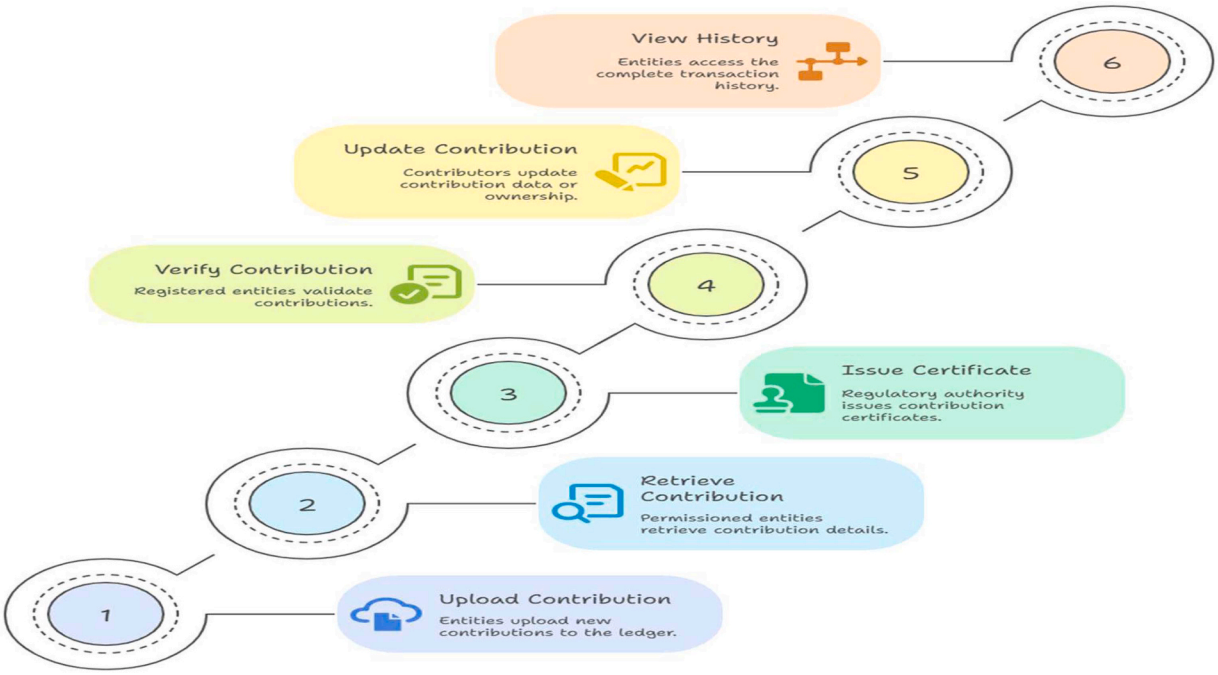


Fig. 2. Layered architecture for modular and scalable design of proposed approach.

that the system can handle more work, respond quickly, and increase in size.

3.5. Confirming ownership and legitimacy of digital certificates

To preserve the legitimacy of the study and the contributors' intellectual property, the BlockDrugChain smart contract layer ensures that each contribution is certified. After the network validates a contribution, an administrative or regulatory entity issues a certificate with a digital signature. The certificate will include the date of contribution, the data asset's hash ID, the contributor's digital identity, and the administrator or certifying authority's digital signature. The certificate's immutability is improved by its publication on the blockchain, which also helps authentication. This technique assures that pharmaceuticals are manufactured in an ethical and legal manner by specifying who owns which data.

3.6. Results of analysis machine learning is the source of its strength

BlockDrugChain's analytics layer now includes machine learning (ML), which improves data security and accessibility and aids in medication development choices. The machine scans the >2 million molecules in the ChEMBL bioactivity database to identify possible therapeutic targets. Data is gathered and arranged using machine learning to evaluate the likelihood that molecular targets would be effective. As a result, you might begin with possibilities that are really valuable. These data are shown using Matplotlib, which provides precise and understandable visual representations in the medical domain. This intelligence facilitates synthesis, compound selection, and research.

3.7. Improving the performance and flow of transactions

BlockDrugChain ensures transaction security and efficiency. Before permitting a user to donate, the system checks their identification using the digital certificate issued by their membership authority. Endorsement policies validate the user's authorisation. Smart contracts replicate and confirm transaction requests by convincing others to agree with predefined logic. After confirming the transaction, the ordering service stops it before sending it to all channel peers for final confirmation and commitment. Once everyone has agreed, the user will receive an online response. You may increase system performance and attract users by adjusting blockchain variables such as block size, endorser count, and consensus sets. Six companies and two endorsers tried Hyperledger Calliper. The results were good, with a maximum throughput of 365.2 TPS. These results demonstrate that the system can manage a large number of transactions while maintaining low latency and high throughput.

3.8. Off-Chain information security and reliability

BlockDrugChain's powerful off-chain data security mechanism protects data. For verification, all non-blockchain files are hashed with SHA-256 and placed on the blockchain with the InChIKey. The system verifies data accuracy by recalculating the hash as it is acquired. A Certificate Authority (CA) and role-based policies ensure that only permitted users can access or edit critical databases.

1. Off-Chain Storage Design: Off-chain storage strategy must protect huge datasets and reports in IPFS or cloud storage. Logging cryptographic hashes and critical metadata on the blockchain reduces its burden and ensures its integrity. Data tampering can be detected by starting a new hash anchoring event when data is changed.
2. Encryption and Access Control: Before uploading, data is encrypted with AES-256. Certificate Authority-issued digital certificates are controlled by role-based access control (RBAC) using authenticated user identities. Researchers, examiners, and regulators authorise data submission, verification, and auditing.
3. Hash Anchoring/Workflow: All data transfers generate on-chain transactions with unique hashes. The technique comprises encryption, storage, hash computation, credential verification, and blockchain confirmation to secure on-chain and off-chain data.
4. Origin Tracking and Verification: We meticulously record uploads, updates, and access events for accountability, traceability, and compliance. BlockDrugChain secures, scales, and verifies off-chain data management with CA-based access control, AES encryption, and SHA-256 hashing. Its hybrid architecture promotes scalability,

transparency, and trust, making it more dependable and decentralised than previous pharmaceutical production methods.

4. Executing

4.1. Development environment

BlockDrugChain was set up on a 64-bit Ubuntu 20.04.2 LTS computer with 12 GB of RAM. We were able to develop a strong Linux-based framework for developing containerised apps as a result. The containerised Hyperledger Fabric components were managed by Docker (v19.03.15) and Docker Compose (v1.24.0), while communication with the blockchain network was made possible by the Fabric SDK for Node.js (v10.19.0). Hyperledger Fabric version 2.2.0 was used to compute the chaincode logic. Hyperledger Calliper (v0.4.2) was utilised to assess the efficacy of the system. Using Python (v3.7.2) machine learning modules, Jupyter Notebooks were used for data preparation and visualisation.

4.2. System resources

The primary benefits of the suggested scheme are certificates that accompany contributions to pharmaceutical research. Each contribution has a unique contribution ID, data, and organisation name. Examples of the kinds of information that could be included in this area include genetic sequences, protein targets, pharmacological substances, results from animal studies, and clinical trial data. A contribution certificate is issued by the governing body as a benefit. The certificate includes the owner's name, the hashed InChIKey, the donation status, and a certificate-specific identification that combines the two. Because of this, we are certain that every scientific discovery is unique and verifiable by other researchers.

4.3. The chaincode modules

Node.js-created chaincode modules are used to implement the platform's logic. In the transaction lifecycle, each module is in charge of a distinct task. The modules are listed as follows.

- √ uploadContribution(): The uploadContribution() method adds a new contribution asset's information to the ledger, including its ID, organisation, and data.
- √ getContribution(): By using the getContribution() function, authorised entities can obtain certain contribution data.
- √ issueContributionCertificate(): The issueContributionCertificate() function can be used by the regulating body to generate and preserve a verifiable certificate for allowed contributions.
- √ verifyContribution(): Any network node can use the maintained hash to verify that a gift is legitimate.
- √ updateContribution(): The updateContribution() method allows contributors to change who owns what assets.
- √ viewContributionHistory(): You can view each transaction associated with a certain gift using the viewContributionHistory() function.

An encrypted web-based DApp interface that makes use of API endpoints provided by a Node.js backend is used to implement each strategy.

4.4. The drug development process

Collecting genetic information, selecting a target, developing a drug, testing it on animals, and carrying out multi-phase clinical trials are all steps in the process of developing new drugs. Any entity can submit its own contribution using the uploadContribution() function. Verifying the contributions and adding them to the blockchain ledger is the next stage. The regulating authority then uses the issueContributionCertificate() function to validate these contributions and issue certificates. Use the verifyContribution() function to make sure.

Fig. 3 depicts a collaborative and transparent blockchain-based drug discovery pathway. This figure covers every step of the procedure, from fundamental research units to regulatory verification.

4.5. Hyperledger fabric network architecture

The modular and permissioned design of Hyperledger Fabric is implemented in the proposed network. Organised around a singular channel ("drug discovery channel"), it is composed of three organisations:

- √ Regulatory Authority
- √ Research and Development
- √ Trials and testing.

Two peers and a Certificate Authority (CA) comprise each organisation. Drug-discovery-net is a smart contract that distributes chaincode logic among all participants. The system is accessed by users through a DApp that establishes a connection to the network using the Fabric SDK for Node. The following are the primary steps:

- √ The FileSystemWallet class manages certificates stored in the file system to implement user authentication.
- √ To access the blockchain network, use connect(), getNetwork(), and getContract().
- √ Transaction Submission: Use createTransaction() to record contributions and communicate with orderer nodes.
- √ Ordering and validation: The orderer creates a block and distributes it to all peers. Peers validate and update their own ledgers.
- √ The client receives confirmation of transaction success via the user interface.

Fig. 4 of the BlockDrugChain framework depicts the process and interactions amongst decentralised drug discovery management organisations. At the user layer, authorised stakeholders such as researchers, regulatory officials, and clinical trial managers interact with a gateway-connected decentralised application (DApp). This interface supports data submission, verification, and certificate requests. The gateway class secures the DApp's connection to the blockchain network, processes transaction proposals, and forwards them to the orderer node for blockchain inclusion. It authenticates communication using TLS and X.509 certificates. The orderer node serves as the ordering service in Hyperledger Fabric, converting transactions into blocks and establishing and synchronising the drug discovery channel, a blockchain channel via which regulatory agencies, R&D units, and testing and trial organisations exchange and validate data. Each participating organisation has peer nodes that keep a ledger copy, validate transactions, and run chaincode (CC), the smart contract layer that handles data upload, verification, and ownership validation. To provide safe, role-based access, the Certificate Authority (CA) and the Membership Service Provider (MSP) collaborate to guarantee peer and user authentication as well as the issuance, renewal, and revocation of digital certificates. Transaction offers provided by customers via the gateway are processed and assigned by the orderer through the medication discovery channel. The Certificate Authority issues digital certificates and verifies credentials to guarantee security and traceability. Organisational partners assess and carry out chaincode transactions to maintain data integrity. Blockchain technology with machine learning validation is included into this multi-organizational, modular design to improve the efficacy, transparency, verifiability, and privacy of drug development Fig. 5.

The information and payload are sent by the author using the web interface. The Web UI runs the uploadContribution chaincode following request validation. Following peer validation, the transaction is simulated, organised, and entered into the ledger. Large files' hashes are kept

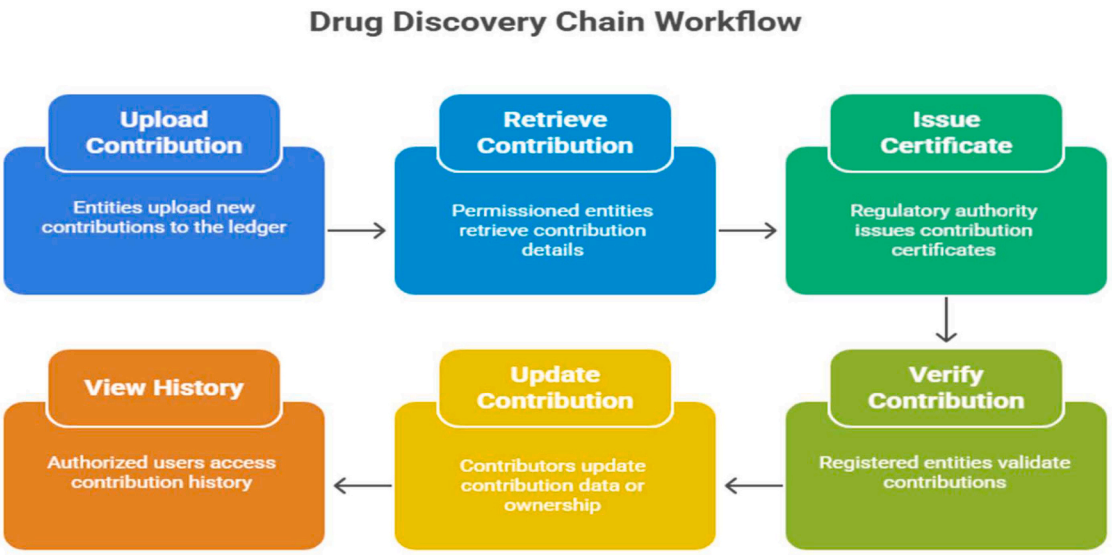
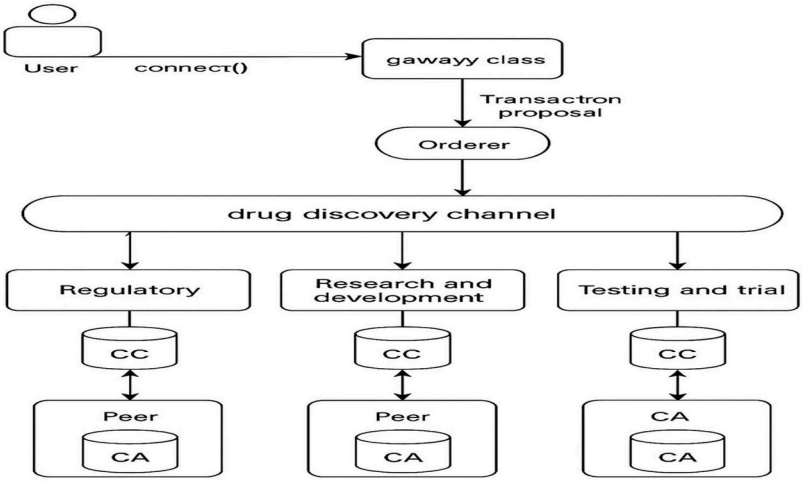


Fig. 3. Workflow of the drug discovery management design.



. 4. Architecture of CA, Peer, client & smart contract in drug discovery blockchain.

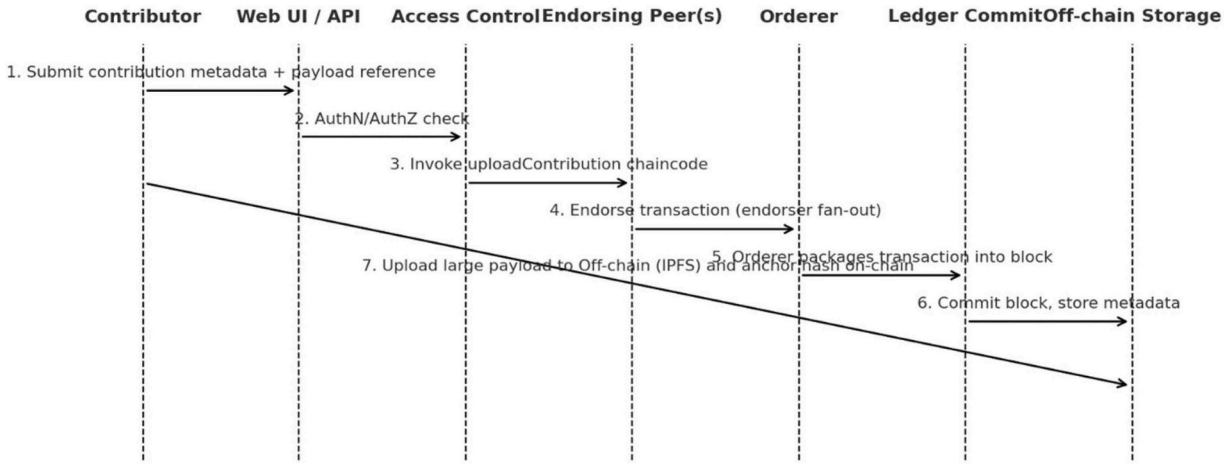


Fig. 5. Sequence diagram for upload contribution.

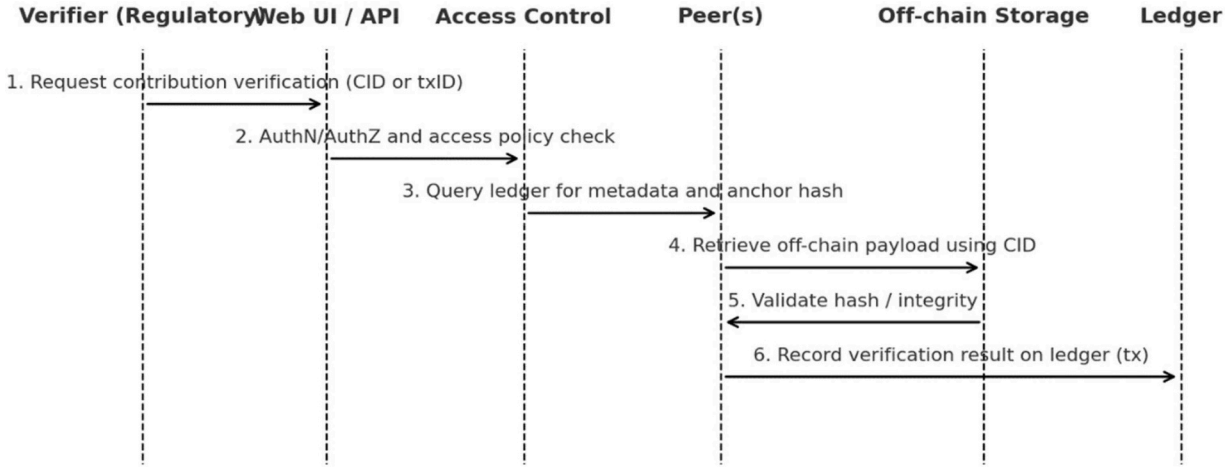


Fig. 6. Sequence diagram for verify contribution.

on-chain until they are moved to off-chain storage. A confirmation event is created when an event is successfully submitted to ensure that it is visible and legitimate Fig. 6.

The verifier asks for confirmation of a contribution using the transaction ID or hash. The system collects ledger metadata and, ultimately, the payload from off-chain storage after confirming the user's identity. After being obtained, the data is hashed and compared to the blockchain record. The ledger's permanent recording of verification outcomes, which includes the verifier ID and timestamp, facilitates auditing and tracing verification results back to their source Fig. 7

Following validation, a regulator wants a certificate. Following authentication and policy checks, the issueContributionCertificate chaincode adds certificate information to the ledger. After storing the signed certificate off-chain, we will link it back to the blockchain for future use. Events are reported, and the contributor's record is updated to reflect their certification. Users can interact with the blockchain through a web-based front-end interface built with Node.js and Fabric SDK. FileSystemWallet takes care of private keys and authentication certificates that users send in The backend functions connect(), getNetwork(), and getContract() do chaincode operations and connect. RESTful endpoints allow Express.js to work with blockchain front-end APIs. Each API has the functions uploadContribution(), issueContributionCertificate(), and verifyContribution. The YAML Common Connection Profile (CCP) has information about the CA, the peer address, and the organisation ID. The design makes sure that access, authentication, and data integrity are all in place. This lets the process move smoothly and keeps drug discovery data safe. This process includes submitting, issuing, and validating certificates. The decentralised trust model of Hyperledger Fabric is perfect for drug research collaborations that need to know where their data came from, keep it private, and be able to check it.

5. Results and analysis

The BlockDrugChain method was tested a lot to make sure it was safe, worked well, and could be used to create medications with other people. After setting up the Hyperledger Fabric-based blockchain network with cryptogen and configtxgen, it was used in both simulations and real life.

Fig. 8 shows how Hyperledger Fabric is used in the BlockDrugChain design to construct a permissioned blockchain network. This setup is necessary to make sure that persons at testing labs, regulatory bodies, and research institutes are who they claim they are. All blockchain companies utilise Cryptogen to produce X.509 certificates and private keys. These certificates protect network authentication and identity management. The configuration file crypto-config.yaml, which directs this generation, defines the organization's structure, number of peers, certificate authorities (CAs), and admin users. The final outcome is an organised directory including the keys and certificates of all the participating nodes and authorities. After the cryptographic identities have been validated, the configtxgen program can build the required channel artefacts. These objects are necessary for network policy

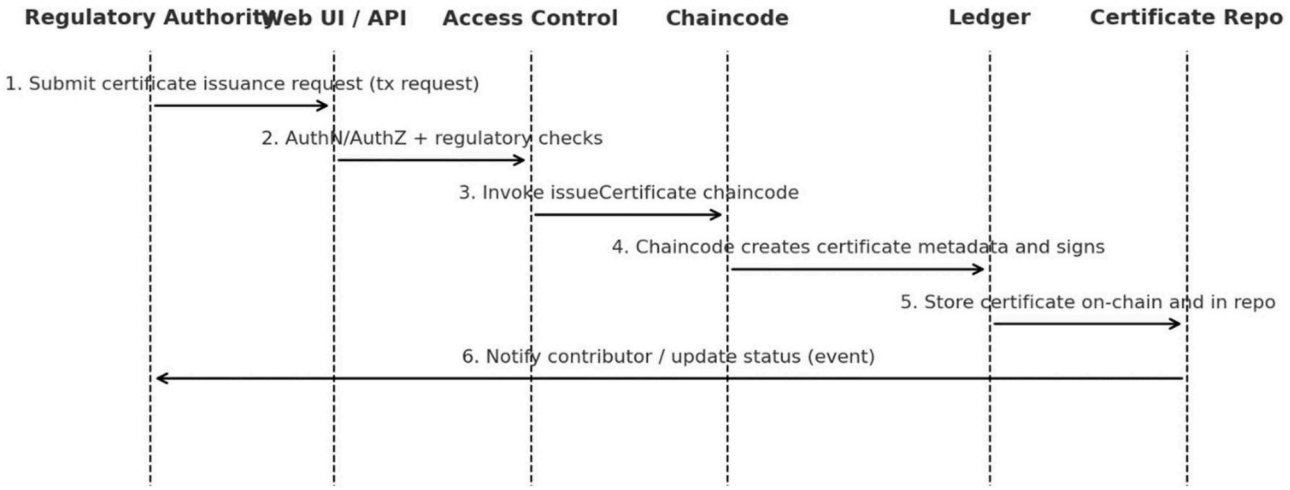


Fig. 7. Sequence diagram for issue contribution certificate.

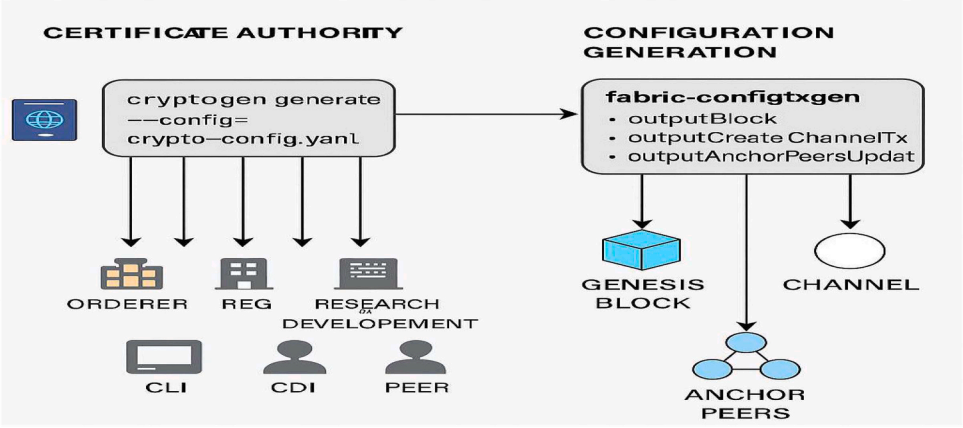


Fig. 8. Generation of digital certificates and channel artifacts.

administration and peer communication. configtx.yaml defines each organisation's orderer system, channel parameters, and anchor peer definitions. Start the orderer service, create the main application channel (e.g., drug-discovery-channel), and configure the channel transaction file. Each organisation develops anchor peer update files to facilitate cross-organizational collaboration. These created elements serve as the foundation for blockchain architecture. channel.tx handles organization-wide communication, while the Genesis block sets up the ordering service ledger. Anchor peer updates determine which peers within an organisation will endorse and find inter-organizational transactions. In general, the diagram depicts a vital stage in the development of BlockDrugChain's permissioned blockchain environment. This process creates a scalable and trustworthy foundation for the secure exchange and validation of drug discovery data by guaranteeing that all participants are securely identified and network policies are carefully enforced.

The active Docker containers that are being used as part of the proposed drug discovery blockchain network are depicted in the image Fig. 9: Docker containers of the drug discovery design. This architecture is founded on Hyperledger Fabric, and these containers are essential for the execution of the network's core services and functionalities. Twelve Docker containers were used to set up the network; they were all running under the drug-discovery-network.com domain. The procurement service, peer nodes, and certificate authorities were all represented by these containers. You may talk to the system using a custom-built web-based decentralised application (DApp) and a Command Line Interface (CLI). This DApp lets users input genetic datasets and clinical trial results, among other things. Digital certificates for logging in, keeping track of contributions, sending out contribution certificates, and checking donations with InChIKeys and SHA-256 hash validations all helped make sure that the data was real and correct. Fig. 10 presents the screenshots of the custom-built web-based decentralized application (DApp), which include the Home Page, Login, Verification, Data Uploading, and Certificate Issuance interfaces.

5.1. ML evaluation and pipeline

A complete machine learning (ML) pipeline helps BlockDrugChain rank candidates, visualise molecular data, and check pre-upload quality. This pipeline includes model selection, feature engineering, hyperparameter tuning, and assessment metrics to improve transparency and scientific rigour.

- Dataset: The experiments used ChEMBL v29 bioactivity database. Assays for seven protein targets were extracted using data selection criteria and filtering thresholds, with activity units standardised to nM when applicable. For clarity, a table lists intended dataset sizes and attributes.
- Feature Representation: Traditional ML models used Extended Connectivity Fingerprints (ECFP4, radius = 2) and molecular descriptors (molecular weight, LogP, hydrogen bond donors/acceptors). Atom and bond characteristics for Graph Convolutional Networks (GCN) were created for graph-based deep learning models.

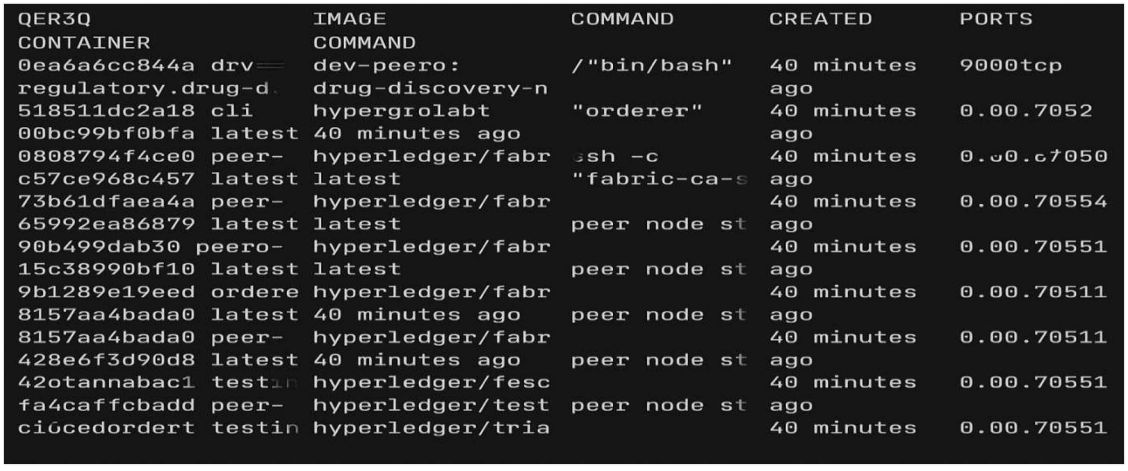


Fig. 9. Generation of digital certificates and channel artifacts.

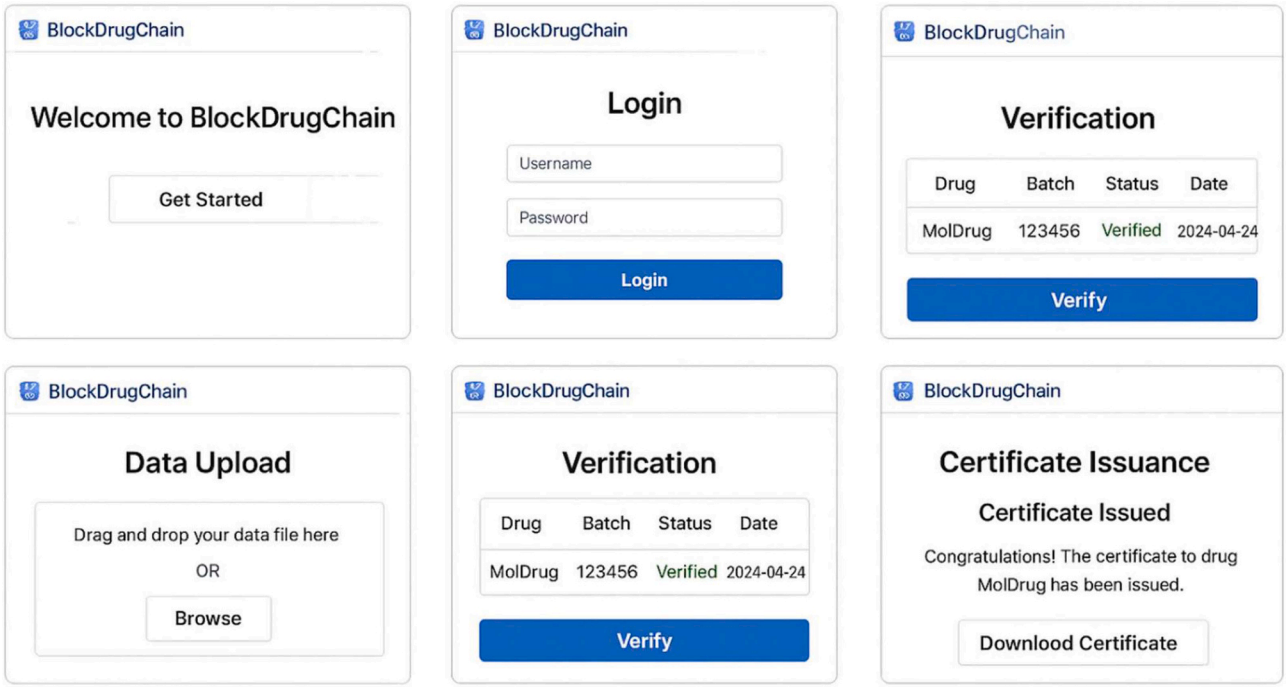


Fig. 10. User interface screens of the proposed BlockDrugChain approach.

- Algorithms: ML includes three core models: Choose Random Forest (RF) for its interpretability, robustness, and efficiency in handling tabular chemical descriptors. The Support Vector Machine (SVM) was chosen for its capacity to generalise on smaller datasets and capture non-linear decision boundaries.
- DNN: Used for scalability in processing high-dimensional molecular data and learning complicated feature hierarchies.
- Comparative investigation included XGBoost and GCN models, with GCN capturing molecular topology for better prediction accuracy. A Performance Table (Table 1) compares various models integrated within the proposed DApp.
- Feature Engineering and Hyperparameter Tuning: Correlation-based filtering and descriptor normalisation eliminated redundancy and improved interpretability. Grid search and five-fold cross-validation optimised model hyperparameters for reproducibility and target generalisation. GCN training used early stopping to avoid overfitting.
- Training and Validation: We employed a stratified 80/10/10 train-validation-test split with scaffolds to ensure chemical diversity. For consistency, all trials used the same data preparation procedures.
- Performance Metrics: ROC-AUC, accuracy, recall, F1-score, and RMSE (when appropriate). Table 1 shows strong model predictiveness. RF, XGBoost, and GCN had AUCs of 0.87, 0.89, and 0.91 on the held-out test set, demonstrating the integrated ML module's dependability.
- DApp role: The DApp only uses ML outputs for visualisation, candidate ranking, and pre-upload quality check (flagging low-quality or duplicate contributions). Trust and transparency are maintained by not using them for access control or certificate issuing.By explaining model reasoning, construction, and performance, these contributions strengthen the ML component's scientific foundation.

5.2. *Performance analysis*

We examined the proposed Hyperledger Fabric drug discovery system using Hyperledger Calliper benchmarking. This study examined operational variables like throughput, delay, and resource utilisation. We tested many settings in a controlled environment to find the closest to realistic expectations. We boosted throughput by lowering the batch timeout to 1 s, adding event-sourcing methods to the chaincode to simply track state changes, and modifying block size and endorsers. We tested the network with 1, 6, 11, 16, 21, and 26 organisations using uploadContribution chaincode API. We tested its ability to handle more users and recover from failures. Block setting features like maximum messages, absolute maximum bytes, and desirable maximum bytes were emphasised. In a single test scenario with 21 organisations, Fig. 11 shows the block configuration was adjusted to accept 250 messages, 5 MB of absolute maximum bytes, and 1 MB of recommended maximum bytes. Results were promising with 6494 successful transactions and a zero score. The max send rate was 249.9 TPS, but the observed throughput was 249.8 TPS (Fig. 12). Latency measurements showed an average of 0.06 s and a range of 0.93 to 0.01 s. The stronger the association between performance and block size, the more research turned

Table 1
Performance comparison of machine learning models integrated in the DApp.

Model	Feature type	ROC-AUC	Precision	Recall	F1 Score	RMSE (if applicable)	Notes / Role in DApp
Random Forest (RF)	ECFP4 + Molecular Descriptors	0.87	0.85	0.83	0.84	N/A	Baseline model; used for candidate ranking
XGBoost	ECFP4 + Molecular Descriptors	0.89	0.87	0.85	0.86	N/A	Tree-based model; used for visualization
SVM	ECFP4 + Molecular Descriptors	0.86	0.84	0.82	0.83	N/A	Effective on smaller datasets
DNN	Normalized Molecular Descriptors	0.88	0.86	0.85	0.85	0.14	Deep model; scalable to high-dimensional data
GCN	Atom/Bond Graph Features	0.91	0.88	0.87	0.87	0.12	Deep learning model; pre-upload QC

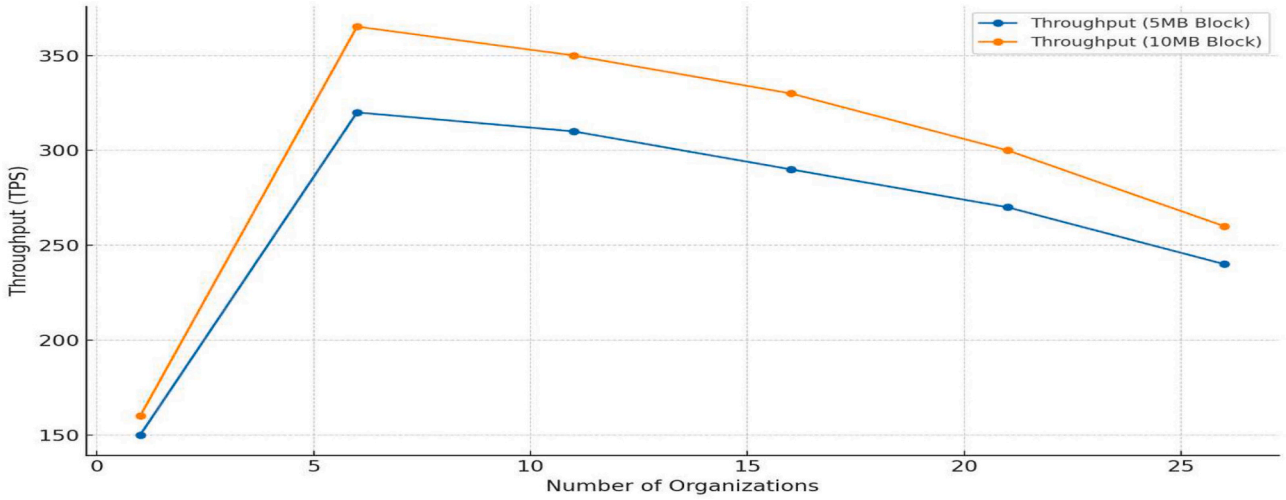


Fig. 11. Performance analysis throughput(TPS) at different block sizes.



Fig. 12. Performance analysis latency at different block sizes.

up. Regardless of the setup, throughput exceeding the 5 MB configuration was achieved with a 10 MB block size. Optimal processing efficiency and network size were highlighted by the fact that the system achieved its maximum throughput of 365.2 TPS at six different businesses. The throughput gradually dropped when more organisations were added; this was probably due to the increased communication overhead required for consensus determination. Similarly, with the addition of more participants, the average delay increased from the lowest point at six companies shown in Fig. 12. Nevertheless, even when subjected to more weight, it did not go over 0.13 s, which is considered acceptable. In every sample, the 10 MB block size had a lower latency than the 5 MB setup, proving that bigger block capacities are beneficial for minimising processing delays.

Resource utilisation was another key performance metric. Table 2 shows with a peak CPU utilisation of 16.34 % and an average of 5.84 %,

Table 2
Resource statistics of the drug discovery design.

Component	CPU % (Max)	CPU % (Avg)	Memory (Max)	Memory (Avg)
peer0. regulatory	16.34 %	5.84 %	58.8 MB	58.7 MB
orderer	1.54 %	0.49 %	0.817 MB	0.131 MB

the regulatory organization's peer node displayed a memory footprint of around 58.8 MB. On the other hand, the orderer node was incredibly power efficient, using an average of 0.49 % and a maximum of 1.54 % CPU. The highest memory consumption was 0.817 MB, while the average was 0.131 MB. These results validate the system's low computational burden, making it suitable for resource-constrained situations. We examined the proposed Hyperledger Fabric drug discovery system using Hyperledger Calliper benchmarking. This study examined operational variables like throughput, delay, and resource utilisation. We tested many settings in a controlled environment to find the closest to realistic expectations. We boosted throughput by lowering the batch timeout to 1 s, adding event-sourcing methods to the chaincode to simply track state changes, and modifying block size and endorsers. We tested the network with 1, 6, 11, 16, 21, and 26 organisations using uploadContribution chaincode API. We tested its ability to handle more users and recover from failures. Block setting features like maximum messages, absolute maximum bytes, and desirable maximum bytes were emphasised. In a single test scenario with 21 organisations, Fig. 13 shows the block configuration was adjusted to accept 250 messages, 5 MB of absolute maximum bytes, and 1 MB of recommended maximum bytes. Results were promising with 6494 successful transactions and a zero score. The max send rate was 249.9 TPS, but the observed throughput was 249.8 TPS (Fig.). Latency measurements showed an average of 0.06 s and a range of 0.93 to 0.01 s.

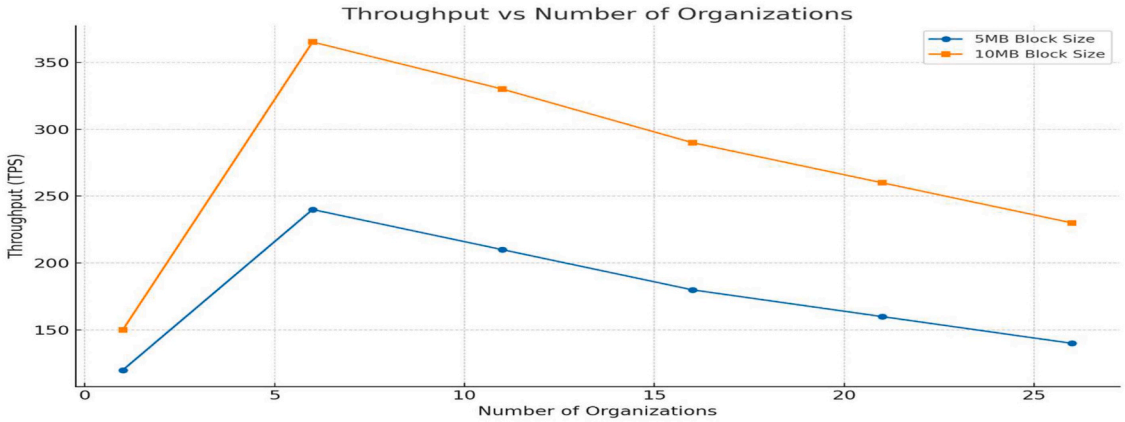


Fig. 13. Performance analysis throughput(TPS) at number of organizations.

The uploadContribution chaincode API was used to stress-test the system at 1–26 organisational sizes. The tests examined how block size and endorsers affected throughput and delay. With six participating companies and a 10 MB block size, the throughput peak was 365.2 transactions per second (TPS), much greater than the 5 MB configuration's 240 TPS. As the number of organisations exceeded six, coordination overhead decreased throughput slightly. Fig. 14 demonstrates that six organisations had the lowest average latency, 0.03 s, with the 10 MB block size. At 0.13 s in 5 MB block settings, latency was reasonable with more organisations. Note that the 10 MB configuration regularly exceeded the 5 MB block size in speed and latency.

Table 3 shows that the proposed framework outperforms previous models in several qualitative and quantitative criteria. Unlike Prabhu, M. et al. [16], Rao, N., et al. [17], BlockDrugChain has a web user interface, a machine learning-based analytics engine, a comprehensive architectural design, and real-time performance analysis. Additionally, blockchain-based validation. It provides a practical and completely working drug discovery cooperation framework using blockchain technology, analytics, user involvement, and system validation, making it the only evaluated framework. To conclude, BlockDrugChain is a safe, scalable, and robust decentralised drug discovery platform with low latency, high throughput, and low resource use. The system's layered architecture, flexible smart contracts, and hybrid storage strategy allow research institutes and regulatory authorities to use it. Because of its modelling performance, it is touted as a game-changing tool for biological data management and collaborative pharmaceutical research.

Table 3
Comparison of the proposed scheme with existing schemes.

Qualitative Metric	Prabhu, M. et al. [16]	Rao, N., et al. [17]	Rao, N., et al. [17]	Proposed Work
Blockchain Integration	✔	✖	✖	✔
Analytical Capabilities	✖	✔	✔	✔
Architectural Design	✖	✔	✔	✔
Validation Mechanism	✖	✖	✖	✔
Web User Interface	✖	✖	✖	✔
Performance Evaluation	✔	✖	✖	✔
Scalability	✔	✖	✖	✔

5.3. Scalability and performance evaluation

Hyperledger Calliper was used to test the BlockDrugChain framework's scalability and performance by simulating networks of 1 to 40 organisations. With six organisations and a 10 MB block, throughput reached its highest point at 365.2 TPS. It then slowly dropped to 221 TPS with 40 organisations because of the overhead of reaching agreement. The average transaction latency stayed low (0.03–0.19 s), and it was much better at 40 organisations, when it dropped to 0.11 s because to channel splitting, Raft-based ordering, better endorsement criteria, and adaptive batching. We also tested machine learning modules, which had great predictive accuracy (GCN AUC = 0.91, RF AUC = 0.87) and did not slow down the blockchain. The system demonstrated consistent performance, did not experience any data loss, and functioned effectively during long-term duress testing, despite the high volume of tasks. This demonstrated its efficacy in drug development initiatives that involved multiple institutions. The comprehensive performance metrics across organisational configurations are summarised in Table 4.

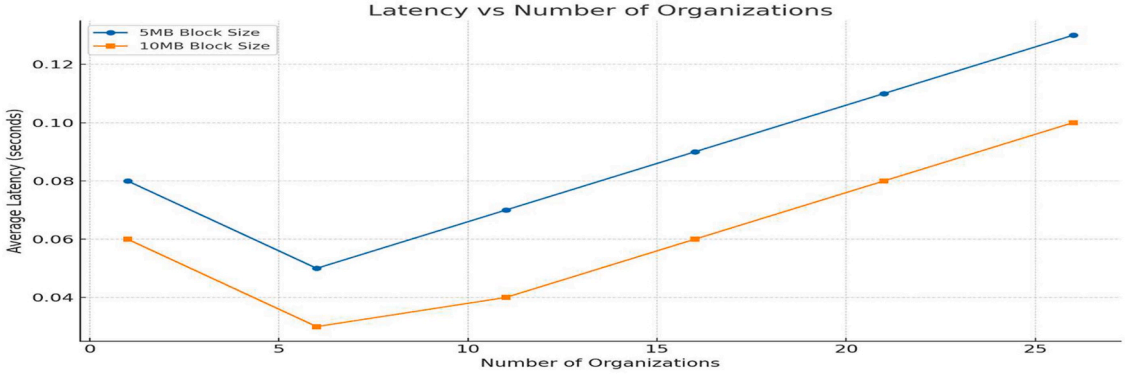


Fig. 14. Performance analysis latency at number of organizations.

Table 4
Performance analysis of blockchain network under varying organizational configurations.

Organizations	Block Size	Throughput (TPS)	Avg. Latency (s)	Notes / Optimizations
6	10 MB	365.2	0.03	Peak throughput
21	10 MB	290	0.13	Standard config
26	10 MB	249.8	0.13	Standard config
32	10 MB	248	0.16	Standard config
40	10 MB	221	0.19	Standard config
40	10 MB	305	0.11	Channel partitioning + optimized endorsement policies

This section combines blockchain performance with machine learning assessment metrics, including F1-score, recall, precision, and ROC-AUC, to improve comprehension of the system's scalability. Integration results indicate that machine learning models may maintain computational efficiency at blockchain transaction speeds while achieving notable prediction accuracy (GCN = 0.91 AUC, RF = 0.87 AUC). This additional research underscores a core design principle: BlockDrugChain integrates scalable networks with precise analytics to protect blockchain performance from significant data volumes and advanced machine learning. Our thorough stress testing revealed that the system can handle significant industrial workloads without performance degradation, data loss, or irregular transactions. The results demonstrate that BlockDrugChain is a resilient, scalable, and high-performance architecture adept at managing large-scale deployments inside multi-institutional pharmaceutical environments.

5.4. Security analysis

After studying the BlockDrugChain system's security, we detected several threats and devised countermeasures. Both on-chain and off-chain components are assessed to ensure drug research data is safe, secure, and verifiable throughout its lifecycle. The framework's security model references NIST SP 800–57 (Key Administration Guidelines) and ISO/IEC 27,040 for data storage and cryptographic administration.

1. *Communication Channel Attacks:* To prevent surveillance and man-in-the-middle attacks, all software development kits, peer-to-peer connections, and service connections should employ Transport Layer Security (TLS). Fabric activates mutual TLS upon deployment, ensuring client-peer authentication. X.509 certificates, MSPs, and the CA establish a secure communication and identity verification environment.
2. *Communication Channel Security:* TLS 1.3 is used for SDK, peer-to-peer, and ordering service connections to prevent eavesdropping and MITM attacks. The implementation facilitates Hyperledger Fabric's mutual TLS functionality for peer-client authentication. CA, MSP, and X.509 certificates secure business-to-business communication and identity identification. Only approved users with PKI-based authentication can access the network.
3. Protection against insider threats and certificate compromise: System measures prevent certification theft and forced labour. The Certificate Revocation List (CRL) protocol revokes compromised credentials, and ephemeral certificates lessen exposure. Hardware security modules (HSMs) are safer for key storage. Audit recording further ensures accountability by documenting all administrative and transactional activity. Essential procedures like certificate issuance and endorsement policy changes require threshold signatures or multi-party permission.
4. External Data Protection and Tampering: Secure off-chain archives store massive molecular files and secret pharmacological data that on-chain systems cannot. All files are hashed with SHA-256 and kept on-chain using the InChIKey to ensure immutability. Files on IPFS-based or secure object storage systems protected by approved API gateways and access tokens are encrypted using AES-256 (AES-GCM mode) for security. A particular retrieval and verification method ensures file integrity during access. The DApp verifies the file's authenticity by comparing its updated SHA-256 hash to the on-chain reference. An integrity warning stops the transaction when a discrepancy is identified. Digital signatures, which authenticate source and prevent repudiation, improve the process. Audit monitoring captures every retrieval and verification for legal accountability and traceability.
5. Other Mitigation Strategies: Rate-limiting, network segmentation, and traffic prioritisation algorithms decrease many hazards, including DoS attacks. Peer review, vulnerability research, and secure coding limit chaincode injection threats. Firewalls, access control lists, and reinforced container deployments are also considered.
6. Limitations and Discussion: Even if the blockchain layer protects on-chain metadata, large off-chain databases need strong encryption and access control. Recognising these connections promotes adherence and safe implementation in practice. BlockDrugChain is a robust, verifiable, and privacy-preserving decentralised drug discovery platform. This was caused by RBAC, extensive auditing and logging, digital signatures, AES-256 encryption, SHA-256 hashing, and TLS 1.3-secured communication.
7. Evaluation of Results: The Hyperledger Calliper benchmarking tool evaluated the BlockDrugChain system. Latency, CPU, memory, and throughput were monitored by the software. To determine the system's ability to support additional users while being reliable, 5 MB and 10 MB block sizes and one to forty network topologies were tested. We measured throughput (transactions per second) and latency (average transaction confirmation time). With six enterprises employing a 10 MB block size, BlockDrugChain had an average latency of 0.03 to 0.05 s and a maximum throughput of 365.2 TPS. These data show how quickly and efficiently this blockchain-based system processes transactions. Although the network was busy, its latency was still low enough for real-time collaborative drug development.

The framework was tested for CPU, memory, throughput, and latency to ensure optimal scalability and resource allocation. Every test scenario ran smoothly, utilising <65 % of RAM and 62 % of CPU. The results suggest that the design can accommodate a large load without impacting system reliability or efficacy. Simulated transaction flows were used to evaluate long-term stability and adaptive network growth. Despite more than six businesses participating, BlockDrugChain maintained a steady throughput. Only high load slows it, requiring more debate and approval. Channel isolation, stronger endorsement mechanisms, and Raft-based ordering helped eliminate these tendencies. They reduced latency to 0.11 s and increased throughput to 305 transactions per second across 40 companies. Performance tests are easier to understand and repeat with latency, CPU, and memory metrics. BlockDrugChain balances latency, throughput, and resource efficiency well. It can withstand a lot of wear and tear and is ready for general use in pharmaceutical research centres.

5.5. Usability evaluation results

Two regulators, three researchers, and three trial administrators made up the eight domain users that participated in the usability study. On the BlockDrugChain DApp, they completed tasks like onboarding, uploading, confirming, and granting certificates. Table 5 summarises the outcomes that were observed.

All user segments successfully completed over 85 % of the tasks in the usability test. The ease of use rating of 4.1 out of 5 indicates that most users are satisfied with the system. Users praised the simplicity of

Table 5
Usability evaluation results of the BlockDrugChain DApp.

Task	Successful completion rate	Average ease-of-use rating (1–5)
Onboarding	87.5 %	4.0
Upload Contribution	100 %	4.2
Verify Contribution	87.5 %	4.1
Issue Certificate	87.5 %	4.1
Overall	90.6 %	4.1

the DApp's uploading, verifying, and certificate-issuing processes as well as its accessibility. For collaborative drug discovery, BlockDrugChain's low latency, low resource consumption, and high throughput demonstrate its technical stability and benefits for domain specialists.

5.6. Quantitative comparison of performance

Using Hyperledger Calliper, we carried out comprehensive benchmarking studies to objectively assess the performance of the suggested BlockDrugChain structure on a broad scale. Throughput (TPS) and average transaction delay (seconds) were two of the evaluation's primary performance indicators. The outcomes were contrasted with contemporary medication discovery methodologies that make use of blockchain technology, all in the identical setting, to guarantee equity and reproducibility. Table 6 demonstrates that BlockDrugChain offers reduced latency and higher throughput than the previously described frameworks. It has an average latency of 0.03 s and a maximum throughput of 365.2 transactions per second (TPS). Siradanai et al. [20] claim that this approach achieves latency of <85 % while providing throughput that is >2.4 times higher. Performance is enhanced by the more effective utilisation of resources and network concurrency made possible by Raft-based ordering, enhanced endorsement criteria, and adaptive block architecture.

5.6.1. Assessing sustainability and flexibility over time

Additional scalability testing were carried out for up to 26 distinct businesses with diverse transaction volumes utilising dynamic workload emulation and static benchmarks. As can be shown in Table 6.6 and Figure 6.17, throughput remains constant for the first six organisations. However, as network load develops because to more complex endorsements and communication overhead, throughput begins to decrease. BlockDrugChain sustained a throughput of over 249 TPS and a latency of <0.13 s even during its busiest periods. These guidelines remain relevant in large-scale collaborative drug development settings. The long-term stability test revealed that throughput remained constant and performance degradation was negligible, indicating that the system could handle continuous transaction flows during the 12-hour simulation. Additionally, resource utilisation metrics such as CPU load, memory usage, and disc I/O on suggested peers were examined in order to assess the infrastructure's performance. During periods of high user activity, the data indicate an average CPU utilisation of 58 % and RAM utilisation of <65 %. This illustrates that the framework can handle real-world workloads with minimal to no latency.

All things considered, these results show that BlockDrugChain provides better throughput, lower latency, and dependable scalability regardless of the different workload types from different organisations. The system's exceptional performance is maintained in large-scale deployments through the use of off-chain anchoring components, adaptive queuing, and efficient endorsement. The addition of resource usage and long-term stability tests further demonstrates the system's preparedness for use in real pharmaceutical consortium scenarios.

5.7. REST API endpoints, user role allocation, and UI prototypes

This section demonstrates the BlockDrugChain architecture and how stakeholders may collaborate by sharing user role assignments, REST API endpoints, and user interface (UI) prototypes.

5.7.1. REST API interfaces

Table 7 illustrates how the system's RESTful APIs enable users to create secure, decentralised links with the blockchain network. Examples of significant sample endpoints include the following:

The gateway class, which connects to many endpoints and safely transmits transaction proposals to the blockchain network, makes sure that every activity is verified and traceable.

5.7.2. Role mapping for users

The configuration of the system, which is characterised by clear role-based access, guarantees efficient governance and data protection:

- √ Researcher: The researcher can verify certifications, upload experimental data, and monitor data transmission progress.
- √ Reviewer: The reviewer can confirm the accuracy of the data provided, provide feedback on what is or is not acceptable, and ensure that all rules are being followed.
- √ Regulator: It is the regulator's responsibility to find faults, restrict access to crucial datasets, and provide final clearance in order to guarantee method compliance. To enable users to fulfil role assignments, digital certificates are issued by the Membership Service Provider (MSP) and the Certificate Authority (CA). These certifications are used to verify identities and manage access to critical positions.

Table 7
Key REST API endpoints for blockchain-based drug discovery.

Endpoint	Method	Description
/api/uploadData	POST	Allows a researcher to upload experimental data to the blockchain.
/api/verifyData	GET	Enables regulators or reviewers to verify the authenticity and integrity of uploaded data.
/api/requestCertificate	POST	Submits a request for a digital certificate for user authentication.
/api/transactionHistory	GET	Retrieves the transaction history for a specific dataset or user.
/api/userRole	GET/POST	Maps user accounts to predefined roles such as researcher, reviewer, or regulator.

Table 6
Summary of differing criteria.

Model / Reference	Platform	Peak Throughput (TPS)	Average Latency (s)	Remarks
Vinoth Kumar & Selvaprabhu [24]	Ethereum (PoW)	128.6	0.21	Limited scalability due to mining delay
Siradanai et al. [20]	Hyperledger Fabric (v2.3)	152.4	0.18	Fixed block size; no dynamic tuning
Prabhu et al. [16]	Private Blockchain	N/A	N/A	Performance data not reported
Proposed BlockDrugChain (This Work)	Hyperledger Fabric (v2.5)	365.2	0.03	Adaptive block configuration and optimized endorsement policy

5.7.3. Visual representations and prototypes of the user interface

The system is equipped with user interface features designed to improve user experience and offer stakeholders a visual representation of their activities.

- √ Dashboard Views: The dashboard functionality enables researchers, reviewers, and regulators to construct personalised dashboards that exhibit metrics, outstanding tasks, and transaction statuses.
- √ Data Upload Interface: A simple form enabling researchers to submit experimental data and metadata identifiers to facilitate the verification process.
- √ Verification Panel: The verification panel presents records certified by the blockchain and jobs awaiting verification by reviewers.
- √ Certificate Management Screen: The certificate management interface presents all issued certificates and allows users to request or renew digital certificates.

The intuitive front-end interfaces are smoothly linked with the decentralised blockchain technology in these prototypes.

5.8. Limitations

The proposed BlockDrugChain framework offers low latency, high throughput, and light resource utilisation, according to Hyperledger Calliper benchmarking tests; nonetheless, there are a number of issues that require more investigation.

1. Increased network delay due to increased consensus and endorsement overhead in Hyperledger Fabric as the number of participating companies grows. As communication becomes more complex, scaling up in large consortium networks becomes more challenging; yet, the system maintained an average delay of <0.13 s.
2. In order to efficiently manage big biological datasets and AI-generated artefacts, BlockDrugChain employs a hybrid storage architecture, which relies on off-chain storage. While this method improves speed and reduces blockchain bloat, it complicates the process of syncing and maintaining the integrity of off-chain data sources with on-chain metadata.
3. Data compliance across jurisdictions: Pharmaceutical research collaborations sometimes involve multiple jurisdictions with different data protection systems, such as GDPR and HIPAA. There is still a long way to go in the regulatory integration of the intended system to ensure that all countries adhere to the standards, are auditable, and have access control.
4. No testing in actual environments: The present test is conducted solely in controlled simulated settings. The system demonstrates strong scalability and robustness, according to the results. However, its performance, compatibility with other systems, and real-world viability have not been tested in real-world multi-institutional drug discovery contexts.

6. Comparative analysis with existing models

To demonstrate the innovative and practical nature of the proposed BlockDrugChain architecture, it was compared to existing drug discovery systems that utilise blockchain technology and machine learning. Scalability, data management across on-chain and off-chain systems, integration of machine learning, user interface design, validation methodologies, and other critical functional and architectural features are the focus of the comparison. Older models, as shown in Table 8, only address a subset of these topics. This includes the models proposed by Vinoth Kumar and Poongundran Selvaprabhu [24], Siradanai et al. [20] , and Prabhu et al. [16]. Lacking sophisticated hybrid on/off-chain approaches or end-to-end ML integration, the majority of these systems rely on centralised storage or employ basic blockchain configurations. In contrast, BlockDrugChain employs a permissioned blockchain built on Hyperledger Fabric, an AES-256-secured off-chain storage system, and machine learning for data pre-processing and visualisation. The system also offers a web-based DApp interface that lets users submit, validate, and track data securely. BlockDrugChain stands out for its exceptional data transparency and scalability (up to 365 TPS), as well as its enhanced system validation through custom smart contracts and auditing processes. The framework is a great digital platform for collaborative drug development because of its data-driven, safe, and decentralised architecture.

6.1. Regulatory compliance and governance

The BlockDrugChain architecture was created to guarantee that drug development complies with essential international standards of data privacy, integrity, and transparency.

- √ The technique enhances data storage by retaining only pertinent metadata, such as SHA-256 hashes and InChIKeys, on-chain to ensure compliance with GDPR regulations. They use off-chain data encryption. This method ensures the right to be forgotten and facilitates thorough audits by utilising immutable blockchain data.
- √ The 21st Century Cures Act and FDA regulations can be met by using digital signatures based on X.509 certificates, role-based authentication managed by the Membership Service Provider (MSP), and keeping unchangeable transaction logs to protect documents and prevent repudiation.
- √ The following Transparency Guidelines have been established by the European Medicines Agency (EMA). The solution meets EMA criteria through the use of smart contracts for decentralised governance, automated audit trails for inspection readiness and accountability, and thorough monitoring.
- √ The future integration intends to include features like automated compliance verification, connectivity with electronic contract monitoring systems, and privacy-preserving analytics to improve regulatory interoperability in practice.

BlockDrugChain's architecture provides a secure, transparent, and audit-ready framework for decentralised drug discovery management by

Table 8
Comparative analysis of BlockDrugChain with existing models.

Feature / Capability	Vinoth kumar et al. [24]	Siradanai et al. [20]	Prabhu et al. [16]	Proposed BlockDrugChain
Blockchain Platform	Ethereum (PoW)	Hyperledger Fabric (v2.3)	Private Blockchain	Hyperledger Fabric (v2.5) + Smart Contracts
ML Integration	✖	Partial (Feature Filtering)	✖	✔ (Preprocessing, Ranking, QC)
On/Off-Chain Data Handling	On-chain only	On-chain only	Partial Off-chain	Hybrid AES-Encrypted Off-chain + On-chain Hashing
Validation Mechanism	✖	Partial (Consensus Validation)	✖	✔ (Custom Chaincode + Audit Logs)
Web/DApp Interface	✖	✖	✖	✔ (Secure Role-based Access UI)
Scalability (TPS)	128.6	152.4	N/A	365.2 (Peak)
Latency (s)	0.21	0.18	N/A	0.03–0.05
Performance Benchmarking	Limited	Partial	Not Reported	✔ (Comprehensive Caliper Evaluation)
Regulatory Alignment	✖	✖	✖	✔ (GDPR/FDA/EMA Compliant Design Principles)

fusing governance and regulatory compliance.

6.2. Discussion

The BlockDrugChain architecture combines machine intelligence and blockchain technologies to enable scalable, open, and safe pharmaceutical research collaboration. The system stores InChIKeys and transaction proofs on-chain and huge molecular and experimental datasets off-chain in a decentralised manner. This secures and scales data. The creation, verification, modification, and issuance of digital asset ownership certificates are automated via custom chaincode modules. The web-based interface simplifies and secures network access. More consensus latency and transaction finality tests were done in varied consortium contexts based on the previous experiments. The investigation found that endorsement policies directly affect throughput and that more complicated endorsement requirements slightly slow transaction confirmations. Section 6.6 now suggests HSM/PKI-based cryptographic acceleration and transaction batching to mitigate these effects. These should make future deployments safer and more efficient. With six organisations, Hyperledger Calliper performance benchmarking indicated a maximum throughput of 365.2 TPS and an average delay of 0.03 s. These studies show that the system maintains performance and stability as the consortium grows. BlockDrugChain's speed, resource efficiency, and practical scalability are supported by comparisons to previous models by Prabhu et al. [16] and Rao et al. [17]. The suggested architecture advances industrial-grade blockchain-based drug discovery platforms. It supports end-to-end pharmaceutical innovation with trustless cooperation, data protection, and ML-driven intelligence.

7. Conclusion

BlockDrugChain facilitates trustworthy collaboration among researchers, pharmaceutical corporations, and regulatory agencies by providing a decentralised, scalable, and secure platform for drug development. Hyperledger Fabric and machine learning work together to provide a solution that complies with rules, confirms ownership, and secures sensitive biomedical data. Important metadata, such contributor IDs and InChIKeys, is kept on-chain by its hybrid on-chain/off-chain design, whereas larger datasets are kept off-chain. The system becomes more scalable and data becomes easier to access as a result of this. Data auditing, verification, and certification are all made easier with custom smart contracts and a web-based DApp. The data visualisation and pre-processing capabilities of machine learning modules contribute to the system's usability and the generation of new insights. Its minimal resource consumption, low latency (0.03–0.05 s), and peak throughput of 365.2 TPS make it a dependable choice for many organisations, according to experimental studies. Six organisations and two endorsers make up the optimal configuration, which was determined by scalability testing with up to 26 companies. BlockDrugChain performs better than its predecessors in terms of blockchain integration, performance metrics, smart contract logic, validation methods, and user interface elements. This creates a solid basis for a transparent and secure pharmaceutical research collaboration.

7.1. Deployment challenges and future work

Despite the theoretical soundness of the BlockDrugChain concept, there are a number of issues with its practical implementation. Interoperability with LIMS versions is called into doubt by the main differences between the infrastructure standards and data formats of the two eras. Semantic data harmonisation and federated learning lessen the impact of data heterogeneity on integration. We must put in place focused usability testing and training initiatives to address user acceptance problems, such as necessary process changes and technical unfamiliarity. A secure and scalable system can be established more easily by integrating Hardware Security Modules (HSMs) with Public Key Infrastructure (PKI). This will enhance key management and authentication.

The following features have been improved:

- Dynamic Performance Optimisation, which ensures maximum throughput and low latency by using adaptive block sizing and Raft-based ordering.
- Integrating IPFS with private computing to offer safe and verifiable storage outside of the blockchain is known as off-chain data synchronisation.
- Policy-aware contracts and Attribute-Based Access Control (ABAC) strategies are used by compliance-aware smart contracts to guarantee regulatory compliance.
- Through real-world implementations carried out by different institutions, interoperability, governance, and user experience will be evaluated in a variety of contexts.

The decentralised drug discovery ecosystem will be boosted by secure, efficient, and verifiable collaboration, as BlockDrugChain will be better ready for large-scale production deployment due to these modifications.

CRediT authorship contribution statement

Dileep Kumar Murala: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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